DGMS 2019 • SECTION I: HYDROGEOLOGY & ENVIRONMENTAL GEOLOGY

Question 72: Physical Determinants of Intrinsic Permeability: Pore Size & Pore Shape

Kozeny-Carman Equation, Hubbert's Fluid Independence, and Microscopic Pore Geometry

OFFICIAL EXAMINATION QUESTION

The intrinsic permeability of the porous medium depends on the

(a) Size of the opening through which the fluid moves only

(b) Shape of the opening through which the fluid moves only

(c) **Both 1 and 2 [CORRECT]**

(d) Neither 1 nor 2

VERIFIED ANSWER KEY

Correct Option: (c) Both 1 and 2 — Intrinsic permeability (k) is an exclusive property of the solid porous matrix (independent of fluid properties). As mathematically formulated in Hubbert's derivation and the Kozeny-Carman equation ($k = C \cdot d^2$), it depends directly on both the **size** of the pore openings (cross-sectional throat diameter squared, d^2) and the **shape** of the openings (tortuosity, angularity, sphericity, and packing geometry represented by dimensionless factor C).

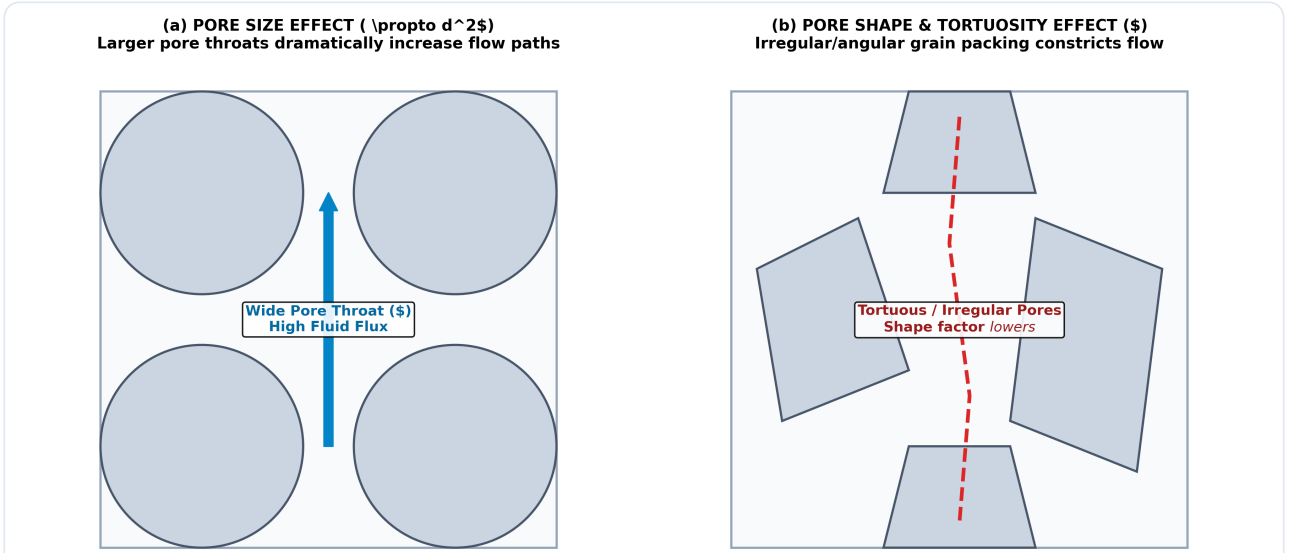


Figure 72.1 | Geometric Controls on Intrinsic Permeability (k): (a) **Pore Size Effect:** Wider pore throat openings (d) reduce frictional viscous drag along mineral grain walls, drastically increasing fluid discharge ($k \propto d^2$). (b) **Pore Shape & Tortuosity Effect:** Angular, platy, or poorly sorted grains create convoluted, tortuous flow paths ($T = L_e / L$), lowering the dimensionless geometric shape factor C and reducing permeability even at identical total porosities.

1. Definition of Intrinsic Permeability vs. Hydraulic Conductivity

In groundwater dynamics, fluid movement through a porous rock or sediment is governed by **Darcy's Law**:

$$Q = -K A \frac{dh}{dl}$$

where K is the **Hydraulic Conductivity** (m/s or m/day).

In 1940 and 1956, **M. King Hubbert** analyzed the microscopic physics of Darcy's law and demonstrated that hydraulic conductivity incorporates both the properties of the fluid and the properties of the rock:

$$K = k \left(\frac{\rho g}{\mu} \right)$$

where:

- ρ is fluid density (kg/m^3)
- μ is dynamic viscosity ($\text{Pa}\cdot\text{s}$)
- g is gravitational acceleration (9.81 m/s^2)
- k is the **Intrinsic Permeability** of the porous medium.

Because k is stripped of all fluid terms, it has dimensions of area ($[L^2]$, expressed in m^2, cm^2, or *Darcies*: $1\text{ Darcy} \approx 0.987 \times 10^{-12}\text{ m}^2$). It depends **strictly on the geometrical properties of the solid skeleton**.

2. Mathematical Derivation: Kozeny-Carman & Fair-Hatch Equations

The classic Kozeny-Carman equation models the porous medium as an assembly of interconnected capillary tubes:

$$k = C_0 \cdot \frac{n^3}{(1 - n)^2 \cdot S_0^2} = C \cdot d_m^2$$

This fundamental equation proves that intrinsic permeability depends on two distinct geometric parameters:

1. **Size of the Openings (d_m^2):** The mean pore diameter or effective grain size (Hazen's effective diameter d_{10}^2). Fluid friction occurs against the pore walls; by Poiseuille's law, flow rate through a channel is proportional to the square of its hydraulic radius. Doubling the pore diameter quadruples the intrinsic permeability!
2. **Shape of the Openings (C):** A dimensionless coefficient governed by:
   - **Pore Throat Sphericity & Roundness:** Smooth rounded grains form open, regular pores; angular grains create tight, irregular crevices.
   - **Tortuosity ($\tau = L_e$ / L):** The ratio of actual curved flow path length (L_e) to straight-line distance (L). Highly tortuous channels reduce fluid flow rates.
   - **Pore Aspect Ratio & Grain Packing:** Rhombohedral vs. cubic packing geometries alter channel cross-sectional profile.

3. Comprehensive Option Analysis

- **(a) Size of the opening through which the fluid moves only:** *Incorrect.* Incomplete because two porous media with identical average pore sizes can have vastly different permeabilities if one has smooth cylindrical pores while the other has highly flattened, tortuous pores.
- **(b) Shape of the opening through which the fluid moves only:** *Incorrect.* Incomplete because pore size exerts a primary quadratic (d^2) control on flow resistance.
- **(c) Both 1 and 2: Correct.** Intrinsic permeability is determined jointly by the size (d^2) and the shape (C, tortuosity, cross-sectional geometry) of the interconnected pore openings.
- **(d) Neither 1 nor 2:** *Incorrect.* Contradicts basic hydrogeological physics.

4. Authoritative Textbook References

- **C. W. Fetter (2001)** *Applied Hydrogeology*, 4th Ed., Prentice Hall, Chapter 4 ("Groundwater Flow: Intrinsic Permeability and Kozeny-Carman Equation"), pp. 84–88.
- **David Keith Todd & Larry W. Mays (2005)** *Groundwater Hydrology*, 3rd Ed., John Wiley & Sons, Chapter 3 ("Hydrogeological Properties: Darcy's Law and Permeability"), pp. 63–70.
- **M. King Hubbert (1940)** "The Theory of Ground-Water Motion", *Journal of Geology*, 48(8): 785–944.
- **Freeze & Cherry (1979)** *Groundwater*, Prentice-Hall, Chapter 2 ("Physical Properties and Principles"), pp. 26–30.